Training Manual for Maize Production in Nigeria



Prepared By:
NCRI, Badeggi
Dr Madukwe Donald
Uke Obinna Dominic
Yusuf Yahaya Mohammad

June 2021

OCP AFRICA

USAID FROM THE AMERICAN PEOPLE

WEST AFRICA TRADE & INVESTMENT HUB

FEED THE FUTURE The U.S. Government's Global Hunger & Food Security Initiative

Editorial Team

Name	Organization
Donald Kelechi Madukwe (Ph.D.)	OCP Africa
Philip Ibrahim (Ph.D.)	NCRI
Uke Obinna Dominic	OCP Africa
Yusuf Yahaya Mohammed	OCP Africa
------	USAID
------	USAID

HOW TO USE THIS MANUAL

This training guide is written for extension workers at different levels of experience. They can use it to train the community knowledge workers/ community-based facilitators, lead farmers, and farmers.

Before the session

Be well organised and read through the training guide before starting a session. It will help you to feel confident. Check that you understand what to do. Organise all the necessary materials for the session like the hand-outs, flipcharts, charts, chairs, and training venue.

Beginning the Session

Greet and welcome the participants. Ask questions about the last session. For example, you can ask:

- What did we learn in the last session?
- What did you apply in your farm that you learned from previous sessions?
- What did you tell your family or community members about the session?

During the Session

Remember to speak audibly and be clear. Pause to allow time for the group to ask you questions.
If farmers cannot see the board or picture during an activity, explain what is shown.

During group work, provide the farmers with flipcharts and markers. Write the participants' suggestions, answers and ideas on a flipchart and supplement with the content provided in the guide. Remind your audience that they can ask you questions if they do not understand something.

Ending the Session

1. Check what your audience learned using the questions in the question and answer activity.
2. Thank your audience for participating.
3. After each module, ask yourself the following questions and write down your answers:

• What went well in the session and why?
• What would I do differently next time?
• Which part of the session did the learners seem most interested in and why?
• Which part of the session did the learners seem least interested in and why?
• Did you manage time well? If not, why?

Reflect on each session in order to find out where you can make changes to improve later sessions.
This manual has been designed to allow readers to select and review specific sections that are relevant to their interests. Therefore, each module and sub-module are self-contained and can be read on its own. Because the focus of the guide is on how to successfully cultivate maize in the field. The manual is divided into the following modules:

Table of Contents

Module 1: Overview	4
Site Selection and Soil Type	4
Soil Test	4
Land Preparation	5
Module 2: Recommended varieties	6
Open-pollinated maize varieties (OPV) recommended for Nigeria	6
Maize hybrids recommended for Nigeria	6
Germination Test for Seed Quality	6
Treatment of Seed Before Sowing	7
Module 3: Planting	9
Time of planting	9
Planting depth	9
Seed rate and plant population	9
Plant population	9
Methods of planting	10
Module 4: Fertilizer Rate, Time of Application and Management Practices	12
Activities during Growth Period	12
Chemical weed control	14
Precautions while spraying in the field	14
Module 5: Growth stages of maize	16
Intercropping and rotations with legumes	17
Yield forecasting in maize	18
Module 6: Management of major disease and insect pests	21
Module 7: Management of Aflatoxins in Maize	24
Module 8: Efficient Harvesting, Storage and marketing of Grains	26
Drying, storage and bagging	27
Storage pests and control	30
Transport	30
Module 9: Record Keeping, Budget Planning & Decision making	32
Record keeping	32
Farm budgets	32
Decision making	33
Appendix	

Module 1

Overview

Key points in this module:

1. Site selection and soil type
2. Soil test
3. Land preparation

Maize (*Zea mays L.*) is one of the most important grain crops in Nigeria. It is the third most important cereal after sorghum and millet. It is produced throughout the country under diverse environments. Successful maize production depends on the correct application of production inputs that will sustain the environment as well as agricultural production. In Nigeria, the demand for maize is increasing at a faster rate daily as feedstock for animal feeds.

Site Selection and Soil Type

- Maize can be grown on a wide variety of soils.
- It performs best on well drained, well-aerated, warm loams and silt loams containing adequate organic matter and well supplied with available nutrients.
- It does not yield well on poor sandy soils and heavy clay soils, except with heavy application of fertilizers, deep cultivation and ridging is necessary to improve drainage.
- In sandy loam soils, good yield could be obtained with increased fertilization and water management.
- Maize can be grown successfully on soils with a pH of 5.0 - 7.0 but a moderately acid environment of pH 6.0 - 7.0 is optimum.
- The optimum temperature for plant growth and development ranges from 30°C - 34°C

Soil Test

It is desirable to check the fertility of your soils every few years (3 - 4 years) for pH levels (how acid or alkaline your soil is), organic matter and plant nutrients. To do so, take a representative soil sample from the field after you have evaluated it in terms of topography, soil texture, drainage, color of topsoil and past management. Where the above features are uniform throughout the field, take a composite soil sample to represent the soil in the field. Where differences are large, then take one composite soil sample from each of reasonably uniform area of the field. The procedure to collect soil samples for analysis is as follows:

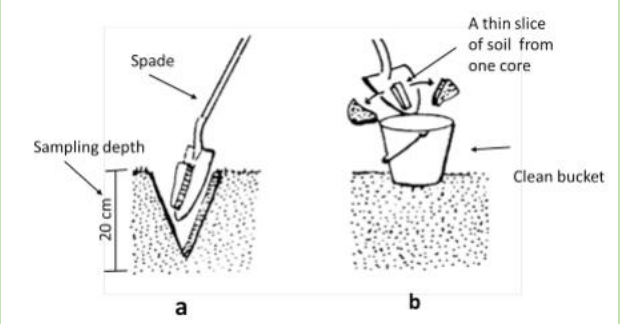


Illustration on how to take a soil sample at a specified depth (a) and a thin slice of one core sample transferred to a clean bucket (b). Figure credit: Frederick Baijukya/IITA.

i. Dig a hole 15-30 cm deep in such a way that a “V” shaped hole is made, cut out a thin slice (approximately 2.5-5.0 cm) and put it in a clean container.
ii. Continue doing the same until you have whole field sampled.
iii. After all the sample cores are taken, mixed the soil thoroughly, and remove any stones or debris.
iv. Dry the soil samples on the clean surface under the shade.
v. In the field, if available, soil testing kit can be used for rapid tests of some soil properties including pH and nutrients (N, P and K) levels.

vi. If test kits are not available, take about 0.5-1 kg of separate soil sample, label it by date of sampling, the farm name and send to the recommended agricultural service laboratories for analysis in order to inform on nutrient deficiency.

vii. Always consult the local agricultural extension agent for advice on when, how and where to analyze your soil samples.

Land Preparation

Land preparation in the savannas is not as rigorous as in the forest zone of the country. In the Guinea savannas, where vegetation cover is fairly dense, land clearing may be carried out by women and children well ahead of the rains. Suitable ridges should be made as soon as the rains start. Although the crop benefits from deep cultivation, yields are not reduced when zero tillage is adopted. For high grain yield, maize should be sown in well prepared ridges 75 cm apart. Because the soil in northern Nigeria is loose, animal-drawn implements can be used for preparing ridges, or these can be made by hand, hoe, or tractor mounted tillage equipment.



Land Preparation. Source: JICA

Summary

- Maize is an important cereal in Nigeria.
- It performs best on well drained, aerated, warm loams and silt loams containing soils.
- Soil tests should be carried out to improve nutrient use.
- Land preparation may be by making ridges with hoes, zero tillage or use of tractors.

Question and Answer Session

1. Is maize an important cereal in Nigeria?
2. What type of soil is not good for maize cultivation?
3. How long is good before conducting a soil test?
4. How do you conduct a soil test?
5. Where is the nearest soil testing laboratory in your location?
6. What is zero tillage?
7. How do you prepare the soil for high grain yield?
8. What equipment can be used for land preparation?

Module 2

Recommended Varieties

Key points in this module:

1. Maize varieties in Nigeria
2. Germination test
3. Seed treatment

Table 1. Open-pollinated maize varieties (OPV) recommended for Nigeria

Ecology	Varieties	Remarks
Southern Guinea savanna	SAMMAZ 15 (IWDC2Syn-F2-W SAMMAZ 40 (DTSTR-Y SYN2 SAMMAZ 51 (IWD C3 SYN/DTSTR)	Use in areas where drought and Striga are problems.
Northern Guinea savanna	SAMMAZ 15 (IWDC2Syn-F2-W) SAMMAZ 27 (EV 99DT-W-STR) SAMMAZ 40 (DTSTR-Y SYN2) SAMMAZ 48 (2011 TZE W-DT) SAMMAZ 51 (IWD C3 SYN/DTSTR)	Use in areas where drought and Striga are problems.
Sudan savanna	SAMMAZ 27 (EV 99DT-W-STR) SAMMAZ 48 (2011 TZE W-DT) SAMMAZ 32 (99 TZEE-STR-QPM)	Use in drought-prone areas and areas with a short season. Where Striga is a problem, use extra-early and drought escape varieties.

Adapted from IITA, 2020

Table 2. Maize hybrids recommended for Nigeria.

Ecology	Hybrids	Remarks
Northern Guinea and Sudan savannas	P48W03	Prolific, combines host plant resistance to Striga and tolerance to Metsulfuron methyl (MSM) for Striga control (4.5 t/ha).
Guinea Savanna	DK777	High grain yield, good stay-green characteristics, and tolerance to *S. hermonthica* (10.9 t/ha).
	SC719	High yield and large grain size (12 t/ha)
	SC651	Tolerance to drought and *S. hermonthica.* High yield potential and good husk cover (9.7 t/ha).
	P48W01	Combined host plant resistance to Striga and tolerance to Metsulfuron methyl (MSM) for Striga control. (5 t/ha).
Guinea savanna and Forest Ecology	30Y87	High yield, excellent stay-green characteristics, uniform ear placement and good standability (12.0 t/ha).
	Oba Super 6	More adapted to Southern Guinea Savanna, Northern Guinea Savanna and Sudan Savanna ecologies, high yield, drought tolerance, low soil nitrogen-efficient, excellent plant, and ear aspect. (7–8 t/ha).

Adapted from IITA, 2020

Germination Test for Seed Quality

- If possible, it would be better to use of certified and disease-free seed to prevent seed born disease, otherwise there is risk of low production.
- Among received seeds from market, the warranty tag of the seed bag should be checked whether that is the right variety or not. After that, the germination test of the seed should be done.

  This activity could be carried out by women and children under proper guidance, so as to facilitate the farming process and save time.

To identify the germination capacity of seeds, the following ways can be followed:

- Keep sand/ soil in a flexible container or carton box, and remove stones from sand/ soil.
- Take 200 seeds randomly from the seed bag.
- Sow the 100 seed in one cartoon box making 10 rows with 10 seeds in each row, and repeat the same process for another 100 seeds in other carton box, for simultaneously getting more exact germination rate.
- During sowing, seed to seed distance should be 2.5 cm. to 3 cm and the depth should be double of the seed size, and should provide daily irrigation.
- Count seedlings up to 7 days to know the vigor of seed. If germination rate of seed was found more than 85%, these seed can be used as seed. But, if it is less than 85% germination, either other quality seed have to be procured or sow the seed more.



Figure 1 Procedure for maize seed germination (Source: JICA Project Team)

Treatment of Seed Before Sowing

Maize seeds should be treated with fungicides for controlling damping off of plant in initial stages. Processes for seed treatment are as follows:

- For seed treatment, 10 g of Apron Star or Marshal Dust is used for 4kg of seed.
- Mix the fungicide with maize seeds in an airtight container and stir vigorously for a few minutes.

The following precautions should be taken when treating seeds with fungicide:

- Windy place should be avoided as it can sweep away powder.
- Children, pregnant women, and old aged people are more vulnerable to pesticides; therefore, they should not be involved while treating seeds, except if they have been correctly trained and they are using the best protective equipment.
- Use protective materials such as masks, gloves and spectacles while treating the seeds.
- Wear clothing that covers the whole body.
- Use pesticides of proper and good quality.
- Wash thoroughly with soap and water after the exercise or when skin comes in contact with the fungicide.

Summary

- Hybrids seeds are recommended because of high yielding capability
- However, the farmer can use open pollinated varieties on his farm.
- Seed germination tests should be carried out to prevent planting bad seeds in the field and reduce supplying of ungerminated stands in the field.
- Seeds should be treated to control crop diseases in the field.
- Safety guidelines must be followed while using chemicals.

Question and Answer Session

1. What maize variety is recommended for your location?
2. Why is it best for your location?
3. Why is it recommended to do a germination test?
4. How is the germination test conducted?
5. Why do we treat our seeds?
6. Why must you use protective equipment when handling chemicals and pesticides?

Module 3: Planting

Key points in this module:

1. Time of planting
2. Planting depth
3. Seed rate and plant population
4. Hand planting
5. Mechanical planting

As most of Maize in Nigeria is rain-fed, early maize is sown in the North by May/June and March/April in the South for Early maturing Seed variety while Late Maturing Maize variety is sown by July/August. Due to erratic rainfall caused by climate change, farmers are encouraged to plant at the onset of rain. There are three major considerations

1. Time of Planting

Early Season

(i) In the Forest zone, the optimum planting date is between 15 March and 1 April, although planting could be done as soon as rainfall becomes steady.

(ii) In the Derived Savannah zone, plant as soon as the rainfall becomes steady, between 1-30 April.

(iii) In the Southern Guinea Savannah zone, planting could still be done as late as May and June, depending on rainfall.

Late Season

1. Late season maize is recommended only for areas where rainfall is likely to be adequate.
2. Popcorn is best planted in the late season where proper drying could be easily affected for optimum popping expansion. Planting should be done preferably between 1–15 August.

2. Planting Depth

Planting and basal fertilizer application is recommended at the same time. The depth of planting should be 3 cm deep and fertilizer is 5 - 10cm deep respectively. Deep seed placement under dry planting is recommended so that seed germinate only after adequate rains have fallen. However, the depth of planting should be uniform to allow uniform plant growth.

3. Seed Rate and Plant Population

Seed Rate

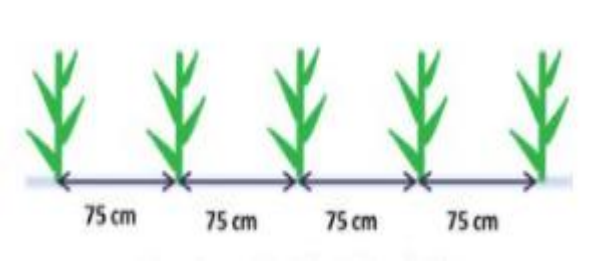
Open pollinated varieties	Hybrids
25 kg/ha	20 kg/ha

Plant 3 seeds per hole and later thin the stand to 2 plants at about 1–2 weeks after germination. When maize is grown as an intercrop, the spacing should be wider than for sole cropped maize. The practice of alternating rows of maize with other crops is highly recommended as it will allow the adoption of recommended packages for each crop.

Plant Population

Rainfall condition	Type of maize	Spacing between rows (cm)	Spacing between plants (cm)	Number of plants per station	Expected plant population /ha
Low rainfall (semi-arid areas)	All varieties	90	60	2	37,000
High rainfall (sub humid- humid areas)	Tall varieties	75	60	2	44,444
	Short varieties	75	20	2	66,666

Plant populations that are higher than the optimum will lead to competition among the maize plant resulting into slender plants that will give low yield. Lower plant population will result into low yields (though with bigger cobs) due to reduced number of ears per unit area. Planting should be planted in rows, without planting in rows, a farmer will never achieve an optimum plant population. In addition, rows ease field operations like weeding and will facilitate harvesting.



Methods of Planting

Planting can be done either mechanically or manually by the hand.

Hand Planting

Hand planting is the most used method in Nigeria. A measuring tape is used to mark 20 cm or 25 cm depending on maize variety along a measuring line for the distances between the seeds. And a second measurement is measured at 70 cm or 75 cm between rows depending on maize variety. A planting stick is used to make two holes at the same time, one at a depth of 5cm for seed and the other at a depth of 10cm for application of fertilizer.



Common equipment used to in conventional land tillage. Photo credit: Africa Conservation Tillage Network (ACTN). a) hand hoes. b) animal traction c) walking tractors d) conventional tractors

Mechanical Planting

This type of planting has the advantage of being quick, and if well supervised will give excellent results. However, if it is poorly supervised, it will give poor-to-disastrous results. It allows you to plant a large area within your pre-determined planting period. Adapt a spacing compatible with other mechanical operations like fertilizer application and weeding. Check the machine well before the anticipated planting date to make proper adjustment. Always read the operator's manual and seek advice from the suppliers for effective usage. Every season make sure that the planter is calibrated to avoid making costly mistakes.

Summary

- Farmers are advised to plant early to take advantage of the onset of rains.
- Seeds should be planted 3 cm deep, with basal fertilizer application 5 – 10cm deep.
- Recommended spacing is 75 cm x 20 cm.
- Planting can be done manually or mechanically.

Question and Answer Session

1. When is the recommended time for planting your maize?
2. What are the three considerations before planting?
3. What is planting depth?
4. Why should you thin your plants to 2 crops per stand?
5. What is plant competition?
6. How do you plant with hand and what are the measurement considerations?
7. What are the advantages of mechanical planting?

Module 4

Fertilizer Rate, Time of Application and Management Practices

Key points in this module:

1. Recommended fertilizer doses
2. Activities during maize growth period
3. Weeding
4. Thinning
5. Transplanting
6. Fertilizer application
7. Chemical weed control and precautionary measures.

The half dose of Urea, full dose of DAP and full dose of Potash should be applied in line and plow the field, then sow seeds in same line. The following fertilizer nutrients are recommended per hectare for maize in the savannas; 100 kg N, 50 kg of single super phosphate (SUPA), and 50 kg of muriate of potash (K20). This recommended fertilizer rate should be applied in two split applications. The first dose (that is 50:50:50) should be applied at planting or within the first week of planting using about 6 bags of NPK 15-15-15. The second dose of N (50 kg, about 2 bags of urea) should be applied between 4 and 5 weeks after planting in the Southern and Northern Guinea savanna zones, but between 3 and 4 weeks after planting for extra-early varieties in the Sudan savanna.



Fertilizer application. Source: JICA

Farmers feel more confident in applying the recommended rates of fertilizer with objects that they identify with such as Coca-Cola caps or using a three-finger pinch.

But because of differences in granule densities of different fertilizers, the facilitator is advised to measure out the recommended weights for each stand of crop in view of the farmers and equate it to a coca-cola cap or three finger pinches. A three-finger pinch is equivalent to 6-gram doses and in the case of NPK 15:15:15 fertilizer which is about a full coca cola cap. With ammonium nitrate fertilizer for instance, a coca cola bottle cap is equal to 4.5 g.

Therefore, assuming a sowing density of 75 cm x 20 cm and 2 plants per stand, a total of 66,666 plants in the field and 33,333 plant stand in a hectare of land. Targeting 6 -7 ton/ha farmer will apply 7.5 bags NPK 15:15:15, farmers should apply 2 caps of Coca-Cola bottle per plant stand, which is equivalent to the recommended 11g per stand. One and half a coca cola cap for urea dosing.

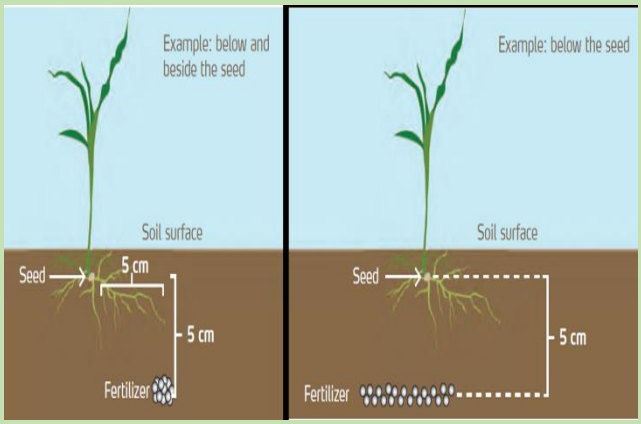
Activities during Growth Period

- We should regularly check the field during 15-20 days after sowing to see if any problems have been occurred in the field.

First farm work: (25-30 days after sowing)

(A) First weeding

- Various unnecessary weeds grow in maize field. Hence firstly, after 25-30 days of sowing, we should weed the field, and take the weeds away from the field.
- Weeding will help the maize seedlings to relieve from competition with weeds.



Placement of basal fertilizers (a) 5 cm below at the side of the seed and (b) 5 cm below the seed. Photo credit: YARA.

B) Thinning

- After 25-30 days of sowing maize or at 4-5 leaf stage, thinning should be done.
- Thinning means practice of keeping only one healthy seedling removing another seedling from each spot.
- Thinning should be done just after the first weeding at the same time.
- Thinning helps to control competition for nutrition, water, and sunlight among the plants. As a result, the remaining plants develops well.

(C) Transplanting

- The removed plants during thinning can be transplanted to the missing spot of field.
- In case the missing spot found, the transplanting should be done after enough watering. But, the root of the maize plant should be pulled out with soil in the root; otherwise, there might be problem of no formation of cob, or no kernel on the cob.

(D) First top dressing

After 25-30 days of sowing, farme have to apply urea immediately as first top dressing. Applying of 3.75 g Urea per stand, first weeding should be done 25- 30 days after sowing. Farmer should apply half Coca-Cola cap of urea near (5-10 cm) to each plant, and cover it with soil to avoid volatilization.

Second farm work: (25-30 days after 1st farm work)

Second weeding: The weeds should also be removed on knee-height stage as second weeding after 20- 25 days of first weeding.



Phosphorus deficiency in Maize Source: Plant disease

(B) Second top dressing

After 20-25 days of first top dressing, during knee height stage, farmer should apply 3.75 g of urea (half Coca-Cola cap of urea) as second top dressing near the plant (5-10 cm from plant) after second weeding, like first top dressing.

(C) Making ridge (Earthing up)

- After the second weeding and top dressing, each nodal root of maize plant should be covered by soil for the prevention of lodging in plants.
- At the same time of covering, we should make the drainage way between each line of plants, which helps to prevent lodging while it rains heavily.

Chemical weed control

Chemical weed control has become more efficient in management of weeds in maize farms. Some recommended herbicides for Nigerian farmers are shown in the table below:

Product	Rate/ha	Estimate for 15-liter sprayer	Condition of use	Remarks
Atrazine	5–6 litres of liquid or 2.5–3 kg of powdered atrazine	400–500 ml	Apply immediately after planting or the following day after planting	Dissolve atrazine powder in clean water before use at the rate of 1 kg in 1 liter of clean water. Thus, the 3 kg should be dissolved in 3 liters of water. This will give about 6 liters of liquid atrazine. Do not apply atrazine if you intend to intercrop maize crop with legume crops (cowpea, groundnut and soybean)
Glyphosate	4 liters	350 ml	Before land preparation	Application rate will depend on formulation and label rate. Wait for 10 days after application before preparing the land.

Other herbicides include: Paraquat at 0.5 kg a.i. ha-1 in 600 litre water. Pendimethalin at 1-1.5 kg a.i. ha-1 is an effective way for control of many annual and broad-leaved weeds.

It should be noted that the actual quantity of water required varies with individuals applying the herbicide due to the following factors:

- Speed of travel or movement of the person performing the spraying.
- Swath width (width of land area covered in one pass).
- Spray pressure of the equipment (force at which the spray is being released).
- Nozzle orifice (size of the opening in the tip of the nozzle in the sprayer).
- The sprayer should be calibrated each time herbicides are to be applied in order to enhance efficacy.

Precautions while spraying in the field

- Read the label on the package before using.
- Only apply chemicals obtained from recognized sources.
- Adhere to manufacturer's instructions and recommendations on the label.
- Use correct kind, dosage and application technique and appropriate safety precautions.
- Determine calibration rate and go by it.
- Seek technical advice when in doubt.

- Do not use expired agrochemicals.
- Wash your hands thoroughly with soap and water after using agro-chemicals.
- Do not eat while applying agro-chemicals.
- Dispose agrochemicals containers in a secure place.
- Wear protective equipment while you apply to ensure safety. You must wear goggles, nose mask, overalls, gloves and closed gumboots.
- If the herbicide accidentally touches the skin and sensitive organs such as the eyes, wash the affected part with soap and clean water. If you feel some pain seek medical attention.



Protective gear while spraying a maize field. Source: PesticideWise

Summary

- ❖ Two coca cola caps (11g) of fertilizer per stand should be applied to maize.
- ❖ Thinning should be done at 25-30 days after sowing maize or at 4-5 leaf stage.
- ❖ Depending on weed density, initial weeding should be done at 25 – 30 days after sowing
- ❖ Care should be taken when handling herbicides and pesticides in the farm.

Question and Answer Session

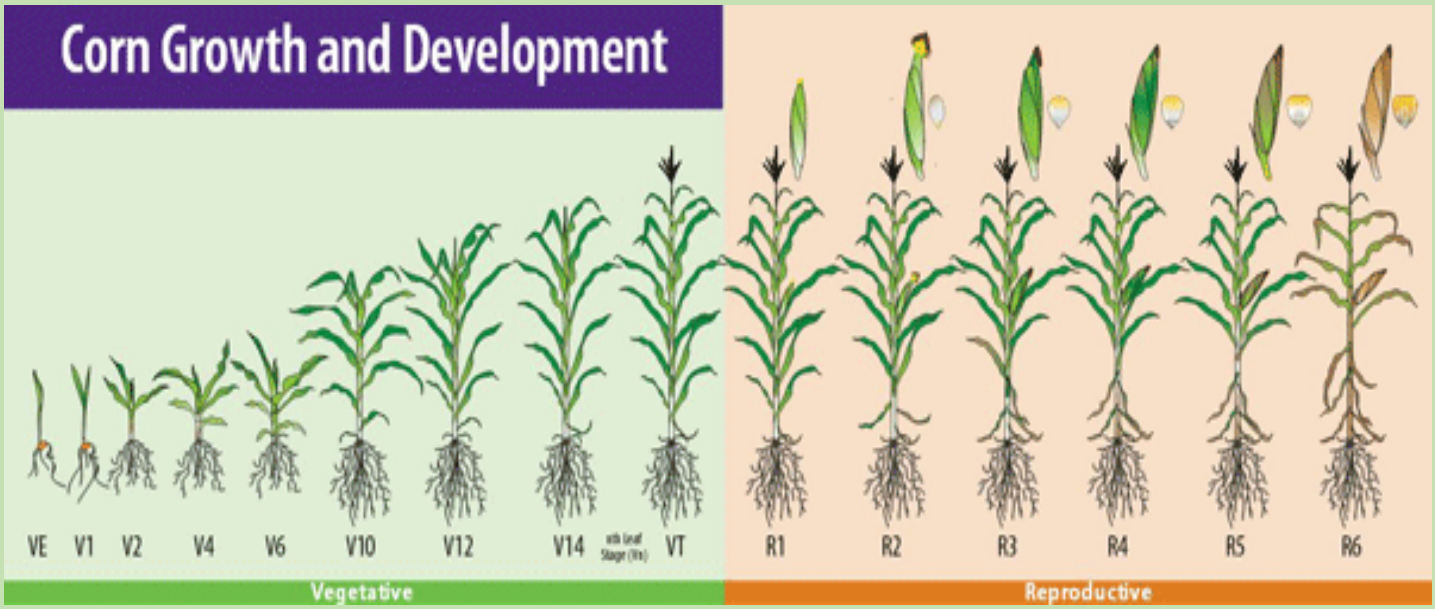
1. How does fertilizer help crop to grow?
2. When is the best time to weed?
3. When is the best time to apply fertilizer?
4. What are the factors that affect herbicide spray and water use?
5. What precautionary measures will you observe before and after handling chemicals?

Module 5

Growth Stages of Maize

Key points in this module:

1. Stages of growth
2. Intercropping and rotation with legumes
3. Yield forecasting



VE - Can occur 4 to 5 days after planting under ideal conditions, but up to 2 weeks or longer under cool or dry conditions.

V1-V5 - At V1, round-tipped leaf on first collar appears, nodal roots elongate. By V2, plant is 2 to 4 inches tall and relies on the energy in the seed. V3 begins 2 to 4 weeks after VE, and plant switches from kernel reserves to photosynthesis and nodal roots begin to take over. Around V4, broadleaf weeds should be controlled to avoid loss. By V5, the number of potential leaf and ear shoots are determined. Plant is 8 to 12 inches tall and growing point remains below soil surface.

V6-V8 - Beginning 4 to 6 weeks after VE, the growing point grows above the soil surface, increasing susceptibility to hail, cold or wind damage. The nodal root system is dominant. At V7, rapid growth phase and stem elongation begin. Number of kernel rows is determined and potential kernels per row begins and continues through V15/16. By V8, the plant reaches 24 inches tall.

V9-V11 - Around 6 to 8 weeks after VE, corn begins steady and rapid period of growth and dry matter accumulation. At V9, tassel is developing rapidly, but is not yet visible. New leaves appear every 2 to 3 days and ear shoots are developing.

V12-V15 - By V12, the plant is about 4 feet tall or more. Nutrients and water are in high demand to meet growth needs. All leaves are full size and roughly half are exposed to sunlight. Brace roots are developing and the potential number of kernels per ear and size of the ear are still being determined. Insect and hail injury can reduce the number of kernels that develop. The plant is about two weeks away from silking at V15. The tassel is near full size, but not visible. Moisture and nutrient deficiencies at this time can reduce the number of potential kernels per row resulting in shorter ears and lower yield potential.

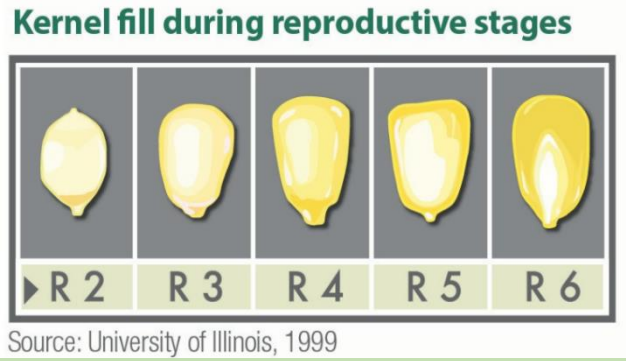
NB: The critical period where maize is most sensitive to stress is 15 days before silking and 15 days after silking (V14 to R2). Bad management during this period drastically reduces yield!

VT - Beginning around 9 to 10 weeks after emergence, corn enters a critical period where successful pollination is required to convert potential kernels into viable, developing kernels. The plant has reached full size. Tassels are fully visible, and silks will emerge in 2 to 3 days. Pollen shed begins and continues for 1 to 2 weeks. Hail can be very damaging at this stage.

R1 Silking - Silking is one of the most critical stages in determining yield potential. For an individual plant it is when the silks are visible. For a field to be in the R1 stage, the average silking date is used. The average silking date is when 50% of the plants have started to silk. Pollination begins at the base and proceeds toward the tip. Potassium uptake is complete, and nitrogen and phosphorus uptake is occurring rapidly. Physiological maturity can be estimated by adding 50 to 55 days to the silking date.

R2 Blister - About 12 days after silking, silks darken and dry out. Kernels are white and form a small blister containing clear fluid. Each kernel develops an embryo. Kernels contain 85% moisture. Stress (especially drought) at this stage can reduce yield potential by causing kernel abortion.

R3 Milk - About 20 days after silking, kernels are yellow and clear fluid turns milky white as starch accumulates. Kernels contain 80% moisture. The effects of stress are not as severe after this stage, but can still result in shallow kernels, stalk cannibalization, or lodging.



R4 Dough - About 26 days after silking, the starchy liquid inside the kernels has a dough-like consistency. Kernels contain about 70% moisture, begin to dent at the top, and have accumulated close to 50% of their maximum dry weight. Stress can produce unfilled or shallow kernels and “chaffy” ears.

R5 Dent - About 38 days after silking, nearly all kernels are dented and contain about 55% moisture. Cob has distinct color: white, pink or red. Silage harvest begins sometime during this stage, depending on desired whole plant moisture.

R6 Black Layer - About 60 days after silking, physiological maturity is reached, and kernels have attained maximum dry weight at 30 to 35% moisture. Total yield is determined. The grain and husks begin losing moisture while healthy stalks remain green. Eventually the leaves will dry off. Harvesting can commence when grain moisture is below 20%. The grain is dried down to 14% for delivery to storage or market

Intercropping and rotations with legumes

- Legumes that are common intercropped with maize are common beans, pigeon peas, cowpeas, groundnuts and soybeans. All legumes fix and add nitrogen to the soil, reducing the need for nitrogen fertilizer for the subsequent crop, and especially so if residues are retained in the field. For example, common beans grown preceding maize, could lead to 5-15 % increase in yields of subsequent maize as a result of improved soil organic matter and nitrogen.



Maize intercropped with Cowpea. Source: Cultivation of crops

- Intercropping or rotation or maize with legumes also helps to minimize disease, pest and weed incidences. When maize and legume like common beans are intercropped, their yields of individual crops are generally lower than when the crops are grown monoculture, but land productivity is usually higher.
- Under intercropping, production costs per unit of output are usually lower and, because legumes sell for up to four times the price of maize, farmers' income is higher and more stable.

Yield forecasting in maize

A pre-harvest yield estimate can be accurate to within 5 percent of the actual harvested yield if the correct procedure is used. When working with trial and demonstration plots, you should always take such a pre-harvest yield sample of both the test plot and the control plot. There is always the chance that the plots might be inadvertently harvested before the agreed-upon time without the yields being measured. Pre-harvest yield sampling is also a quick way of estimating crop yields in farmers' fields.

General Principles of Yield Sampling

- Samples should be collected at random for various portions of the field or plot. Do not purposely select samples from higher- or lower-producing areas within the plot or your estimate may be very inaccurate A random sampling pattern should be determined before you enter the field so you will not be tempted to choose them by visual appearance.
- Don't collect yield samples more than one week before the actual harvest.
- When taking each sample, the area (or row length) to be harvested must be precisely measured. Do not estimate' Remember that any error in the sample area size will be magnified hundreds of times when converting the yield to a larger land unit basis.

- You must adjust the sample weights to account for factors like excess moisture, damage, and foreign matter.

How to Take Samples and Estimate Yields

1. The Sampling Procedure

- Number of samples: For plots less than 0.5 ha, take a minimum of five samples. For plots of over 0.5 ha, take between five and ten. If crop growth is not very uniform, take ten samples.
- Size of each sample: Take each sample from the same sized area or same amount of row length. Individual sample size should be between 2.5 and 5.0 square meters. For row crops, the area of a sample is determined by multiplying row length by row width. (Harvesting three meters of corn row planted in rows one meter wide will give you a sampling area of three-square meters.)
- Taking a random sample: Decide on the sampling pattern before entering the field, and do not deviate from it. To randomize, the field can be divided up into sections and each section given a number drawn from a hat. Or you can pick randomized starting points at the side of the field and then enter random distances from the starting point. A good system for row crops is to number the rows and select them at random, then select the distance into the row (field) at random. NOTE: Exclude three meters or four rows of perimeter from your sampling area along all four sides of the plot to ensure sampling from the heart of the plot.

2. Accuracy: Use a tape to measure each sampling area or row length. Use an accurate scale to record the total weight of the samples within one plot.

3. Handling the Samples: The samples should be harvested and processed according to local prevailing methods. If drying is required before shelling or threshing, be sure the location is secure and free from rodents or birds.

4. Weighing the Sample: Use an accurate portable scale. You do not need to weigh individual samples separately, but only the fatal collective sample from the plot. If you cannot find a good portable scale, have the grain weighed in town.

5. Checking Grade: Take a random sample of the collective sample and have it checked for moisture content and any other graded qualities if necessary. (Refer to the storage section in Chapter 7 for how to determine grain moisture content.)

6. Yield Calculations:

Size of total sample area = No of samples x size of individual sample areas

Estimated yield = weight of total sample area x $\frac{total\ size\ of\ the\ plot}{total\ sample}$

6. Correcting for Moisture: Yields are usually based on grain that is dry enough to store in shelled form (usually 13-14 percent moisture content). If you base your estimates on the weight of a high moisture sample, you should revise the yield downward using this simple formula (otherwise, dry the grain first).

Grain weight after drying = [% dry matter before drying / % dry matter after drying] x original grain weight

Example: Suppose you weigh a collective sample of "wet" grain and then estimate the plot yield to be equal to 3500 kg/ha.

A moisture test shows the sample has 22 percent moisture; what is the actual yield based on 13 percent moisture?

22% moisture = 78% dry matter,

13% moisture = 87% dry matter

$\frac{78\%}{87\%}\ x\ 3500\ kg/ha\ =$ 3138 kg/ha yield based on 13% moisture

A Yield Estimate Example

Suppose you are taking a yield estimate on a farmer's maize plot which is slightly less than 0.5 hectare. The rows are planted 90cm apart, and you decide to take six samples, each consisting of 1/1000th hectare of row length. The collective weight of the shelled, dried maize is 18 kg. What is the estimated yield on a per hectare basis?

Solution:

area of collected sample = 6/1000ths of a hectare = 60 sq. Meters

$18kg\ x\ \frac{10000m2}{60m2} =$ 3000 kg/ha estimated yield

Summary

- It is important that the farmer understands the various growth stages of maize so that he can properly plan fertilizer management for his field.
- Plants should not be deprived of nutrients and water before and after silking. Intercropping maize with cereals helps the farmer improve nitrogen use and reduce risks. Yield forecasting helps the farmer to determine yield in his farm.

Question and Answer Session

1. At what stage is management very important to improve yield?
2. What is silking?
3. When does the kernels start to develop?
4. When is the maize fully developed?
5. What crops do you intercrop with your maize in this location?
6. What are the benefits of intercropping?
7. What are the benefits of crop rotation?

Disease and Cause	Symptoms	Occurrence	Management
Gray leaf spot (GLS)  A foliar fungal disease caused by fungal pathogens *Cercospora zeae*-maydis and *Cercospora zeina*.	Symptoms develop on lowest leaves first and progress upward. Pinpoint lesions surrounded by a yellow halo appear first. These elongate into pale brown or gray rectangular lesions 0.2 to 2.0 inches in size. Entire leaves may become blighted.	The disease survives in infested debris. Initial infections occur in early in one to 2 months in the season. Cloudy weather, accompanied by prolonged periods of leaf wetness and high humidity favor disease development. Severe damage often occurs in low spots or in fields bordered by trees or streams where air circulation is poor.	Use a rotation long enough to eliminate corn debris or use tillage. Many partially resistant hybrids are available. Use fungicides when the economic threshold is exceeded.
Head smut ***Sphacelotheca reiliana***  Caused by soil borne smut fungus *Sporisorium reilianum*	Tassels and ears of infected plants are often replaced by smut sori. On tassels, individual spikelets may be infected. Ears are usually entirely aborted.	Infection occurs in seedlings and the fungus grows systemically into the reproductive parts of the plant. Temperatures of 70 to 82 degrees Fahrenheit and low soil moisture favor infection.	The disease is effectively controlled by resistant hybrids. Planting early and use of systemic fungicide seed treatments are partially effective.
Fusarium ear rot Caused by soil borne fungi *Fusarium verticillioides* 	Usually, individual or groups of infected kernels are scattered on the entire ear. A whitish fungal growth on kernels is typical. Infected kernels often exhibit a "starburst" pattern, i.e. white streaks radiating from the point of silk attachment at the cap of the kernel.	The disease is most common in continuous maize. *F. verticillioides*, produce large numbers of microconidia and macroconidia on crop maize debris and these asexual spores comprise the most important inoculum for Fusarium ear rot. Infection takes place through the silks.	Balanced moisture and fertility and control of insects that damage Kernels reduce the severity of the disease.
Maize leaf blight Caused by the fungus *Exserohilum turcicum* 	Gray green, elliptical or cigar-shaped lesions that are 1 to 6 inches long appear on oldest leaves first and progress upward. Lesions become tan as they mature.	The disease is most common in continuous maize where crop debris remains on the surface. Temperatures of 65 to 80 degrees Fahrenheit and extended periods of dew favor infection.	Resistant hybrids. Use rotation or tillage to eliminate crop debris.

Source IITA, 2020

Pests

Maize stem borer: The maize stem borer, *Busseola fusca*, is a serious insect pest of maize and causes enormous crop losses. Stem borer infestation is higher in the later part of the cropping season.

Symptoms: Maize plant parts affected are leaves, stems and cobs. First instar larvae feed on young terminal leaf whorls producing characteristic patterns of small holes and "window panes" (patches of transparent leaf epidermis) where tissue has been eaten away (A, B). Later, they eat into the growing points which may be killed so that the dead central leaves form characteristic dry, withered "dead hearts". Older larvae tunnel extensively into stems and eat out long frass-filled galleries, which may weaken stems and cause them to break (C). Some larvae bore in the cobs. When larvae feed on grains in the cob, frass deposits are visible (E, F).



Symptoms of maize stalk borer, Busseola fusca, damage on leaves. Photo credit: ICIPE.

Cultural control: Stem borers multiply in crop residues left on the field after harvest. So, turn all the crop residues under the soil immediately after harvest. This practice helps to kill the pupae left in the residues thereby preventing the carry-over population and consequently limiting the establishment of the pest on the following season's crop. Trap crops e.g. elephant grass or early planted maize, reasonably spread around the proper maize field, will entice moths to lay their eggs before the main crop is up.

Chemical control: Spray Karate at 30 ml per 15 liters of water for every 100 m2 beginning at 2 weeks after planting and repeat at 2-week intervals till tasselling stage.

Fall army worm (FAW) *Spodoptera frugiperda*): Almost 100 different crops and other plants are susceptible to attack, but there is a preference for maize, rice, sorghum and sugarcane. In young plants, the stem may be cut. Older larvae stay inside the funnel and so are protected from spray applications and predators. In older plants the larger larvae can bore into the developing reproductive structures, such as maize cobs, reducing yield quantity and quality.



A maize plant attacked by fall army worm. Photo credit: CABI.

Ampligo (Lambda cyhalothrin 50g/kg + Chlorantraniliprole 100g/kg) and Fawligen has been tested and evaluated by the International Institute of Tropical Agriculture (IITA) for two consecutive years in Nigeria. Having access to a safe and effective biological control for FAW will be a welcome relief to many of the Nigerian farmers affected by FAW".

Termites: In areas where termites come early in the growing season, pesticides should be used in the form of seed dressing. Or applied to the planting hole. Insecticides such as Fipronil give some success either as a seed dressing, or as a furrow treatment at planting. In areas where termites come late during the growing season apply pesticides like Termidor, Ostrinia or Imidachloprid as a high-volume spray to the plant bases at the appropriate time (when attack is likely). Termite hills in the field and surrounding area should be located and destroyed.

Corn Earworms: For green maize production in particular and increased grain yield, apply 3 sprays of Cymbush 10 EC of Ambush 25 EC at the rate of 20ml/10 litres of water at weekly interval, beginning from 50% silking. Direct spray to the cobs.

Summary

Maize diseases and pests management includes cultural, biological and chemical methods to fight against their spread.

Question and Answer Session

1. What diseases of maize do you know of?
2. How do you identify grey leaf spot?
3. When you plant maize continuously in a plot of land, what is the most common disease?
4. What pests of maize do you know of?
5. How can you control fall army worm?
6. How do you control termites in a maize farm?

Module 7

Management of Aflatoxins in Maize

Key points in this module:

1. Harmful effects of aflatoxin
2. How aflatoxins are produced

Aflatoxins are poisons produced by the fungus (mould) Aspergillus flavus and closely related moulds. They contaminate 25%+ of maize produced in Sub-Saharan Africa. It poses health risks to both humans and animals even at low concentrations.

How do aflatoxins harm us?



- Aflatoxins cause liver cancer, suppress the immune system, retard growth and development of children, among other health problems.
- Consumption of aflatoxin-contaminated food and feed decreases productivity in humans and animals; sometimes it is fatal.
- Aflatoxin-contaminated crops typically are forced into low-value markets.
- Aflatoxins damage our health and reduce both crop value and business opportunities.

Can we see aflatoxins?



- Aflatoxins are colourless chemicals and cannot be seen.
- Colour of Aspergillus flavus is yellow/green.
- Grains without visible signs of A. flavus may contain high amounts of aflatoxins.
- Aflatoxin concentrations are determined with laboratory tests.

Where and how are aflatoxins produced?

- Aspergillus moulds produce aflatoxins when infect grains before and/or after harvest.
- Insect damage increases fungal growth and aflatoxin contamination.
- Drought and high temperatures increase aflatoxin production as do improper pre-harvest, post-harvest, and storage practices.

Can we reduce aflatoxin contamination?

- Yes, by using aflasafe, a highly cost-effective, environmentally safe biocontrol product developed by IITA and partners. Aflasafe products should be used along with other management practices.

Effect of aflatoxins on health

- Aflatoxins are linked to: cancer (especially of the liver), immune-system suppression, child stunting, impaired food conversion, kidney toxicity, and when consumed in large quantities, can cause death.
- Exposure to aflatoxins may start before birth; mothers can pass aflatoxins to the fetus through the umbilical cord.

What is aflasafe and how does it work?

- Aflasafe products contain four atoxigenic *A. flavus* strains native to the target nation.
- Roasted, sterile sorghum grains serve as carrier and nutritive source for the atoxigenic strains.
- Spores of the strains are coated onto sorghum using a polymer sticker. A blue food dye is used to distinguish aflasafe from regular sorghum.
- Aflasafe is broadcasted on fields at a rate of 10 kg/ha. In maize, aflasafe is applied 3 weeks before flowering.
- Within 2-3 days, atoxigenic strains sporulate on sorghum and displace aflatoxin-producing fungi.

Use Precautions

Do not apply with fertilizers, insecticides, or fungicides. Apply aflasafe only when the possibility for drift to potential sensitive areas (e.g., residential areas, water bodies, known habitat for threatened or endangered species) is minimal.

Summary

Aflatoxins are poisons produced by fungi on maize, and it has serious health implications on our body. Aflasafe is a safe method of controlling it. Efforts should be made to follow safe use of aflasafe.

Question and Answer Session

1. What are aflatoxins?
2. How does aflatoxins affect our health?
3. What is the colour of aflatoxin?
4. How do we control aflatoxin?
5. What is aflasafe?

Module 8:

Efficient Harvesting, Storage and Marketing of Grains

Key points in this module:

1. Harvesting of maize
2. Hand shelling
3. Drying, storage and bagging
4. Storage pests and control
5. Transport

When maize is to be consumed fresh, harvesting should be conducted when the silk has turned brown (50–70 days after planting). But when grain is needed, harvest as soon as the grain is dry enough (80– 110 days after planting) depending on the variety. During harvest, the cobs can be broken by hand from the plant or the whole plant can be cut with a cutlass. Where the entire plants are harvested, they are often stacked in the field to allow the grain to dry further. In the humid zones, the crop can be dried in a traditional ventilated granary.



Allow maize to dry properly before harvest

Postharvest processing: Maize shelling can be done by any one of the following methods:

- Hand shelling
- Use of internally ribbed tubes, beating cobs with a stick.
- Using a single intake disc (hand-operated).
- Using double intake disc (maybe hand/pedal/engine- operated). The method to use is determined mainly by output, although engine operated multi-crop threshers used on a contract basis are gradually becoming popular.



Farm workers threshing maize at Makarfi Local Government of Kaduna State Nigeria. Credits: Wikimedia Commons

Grain cleaning can be done using the wind for winnowing or with screens of proper size.

Farmers using hybrid maize should not save their own seed for their next crop, as the advanced generation hybrid seeds are likely to yield reduction of 25-30 per cent. However, can save seed from composites and open pollinated varieties, when grown in isolation. At least seeds from 500 to 100 ears of the best yielding and normally spaced plants resistant to prevalent diseases and pests should be bulked. Ears should be dried, shelled and treated with an insecticide and the treated seed is necessary, as the untreated seeds on ears are times badly attacked by stored grain pests and the germination is markedly reduced.

Drying, storage and bagging

Maize reaches physiological maturity between 130 and 160 days after planting, depending on the variety. At this stage the crop has a moisture content of about 30 percent, and can be harvested provided that adequate drying facilities are available.

Reasons for drying

- Maize is physiologically mature, that is it ceases to grow and accumulate more dry matter, when the grain moisture content ranges between 30 and 35 percent. At this stage, the grain is still a living organism and, as such, it respires.
- Respiration causes the production of heat, carbon dioxide and water. The higher the temperature and moisture content, the higher the respiration rate and the higher the loss of dry matter.
- The ideal moisture content for storing maize is below 13 percent. For every 1.5 percent of moisture content above that grain deterioration doubles.

On-farm drying

An appropriate drying facility for use by small farmers should:

- Enable the farmer to avoid field drying by permitting adequate drying of maize with a starting moisture content of around 30 percent.
- Utilize locally available building materials where possible.
- Protect the maize against termites, rodents and other animals.
- Protect the crop against rain.
- Not require too much skill to construct.

Two simple methods that can be used to sun-dry shelled maize after it has been partially dried on the cob are:

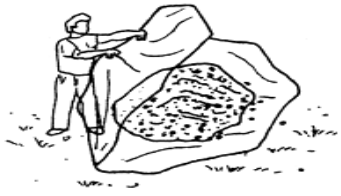
i. Drying on a plastic sheet on the ground.



1 Spread shelled maize no more than 5 cm deep on the plastic sheet



2 Rake the maize from side to side while it is drying



3 Protect the maize at night by folding the sheet in half

Sun-drying shelled maize on a plastic sheet on the ground

ii. Drying in a drying tunnel.

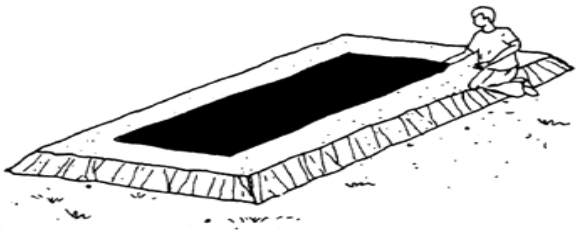


4 You can also gather the edges of the sheet to form a sack

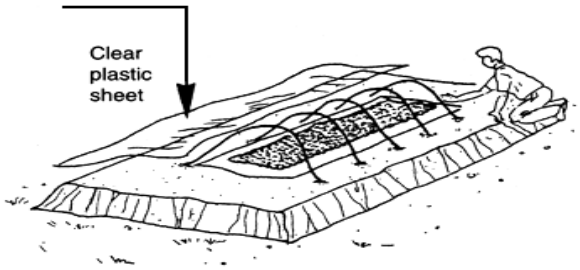


5 Tie the sack and carry the maize inside at night

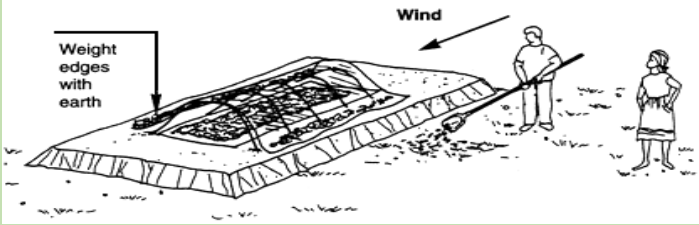
Build a mound of hard-packed earth and lay out a sheet of black plastic to dry your maize on



2 Make a frame of wire, wood or bamboo and prepare a clear plastic sheet large enough to cover it



Sun-drying shelled maize in a drying tunnel



3 Spread shelled maize no more than 5 cm deep on the black plastic, cover the frame with the clear plastic and weight edges with earth

Storage

A store serves the following purposes. It provides protection against:

- Ground and rainwater.
- Insect and other pests
- Excessive heat.

The drying and storage crib, and the mud or cement-plastered basket are examples of the more traditional storage common to the region. The brick bin and the ferrocement bin are examples of improved storage. While improved storage is better, it is also more costly. If improved storage is beyond the reach of the small-scale farmer, traditional methods can be improved upon. Some photographs of these kinds of storage are shown on the following pages



Storing maize in a crib on the ground



Open-basket storage



Maize crib roofed with thatch



Bins roofed with thatch

Bins roofed with thatch



Bins roofed with brick

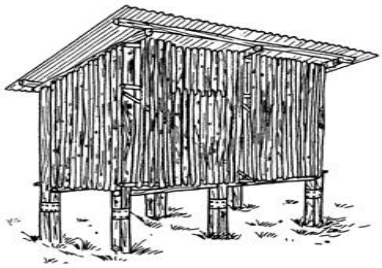
Forming a brick grain chute



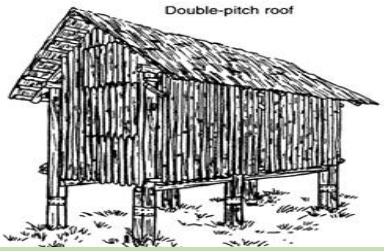
Beginning to inset top bricks to form roof

Examples of rectangular drying and storage cribs

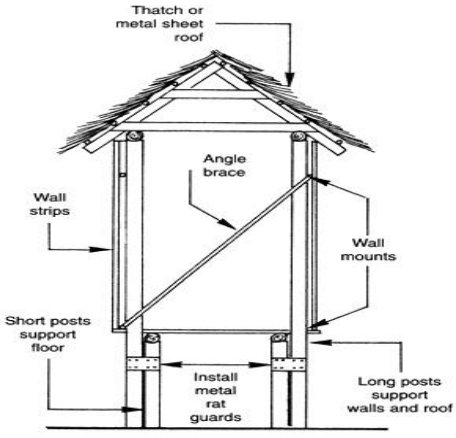
Single-pitch roof



Double-pitch roof



Note: Angle braces are needed between each of the long posts to insure stability of the crib (see section).



2 Section through a drying and storage crib

The major storage pests

Grain weevils (*Sitophilus spp).* Infestation may be by insects already present where maize grain had previously been stored, or by cross-infestation between stores, or by cross-infestation between granaries during storage. In some cases, an infestation may start on the field, especially when maize is left for too long before being harvested. Apart from the hygienic condition of the granary, chemical control measures can also be used.

- Maize that will be stored for 6 months or more should, in addition to hygiene in the store, be treated with phostoxin. Phostoxin tablets or pellets may be used to store more maize in an airtight container or large storage in sacks. Large-scale storage in bags such as pyramids will require a complete cover with tarpaulin after application with 4–6 tablets or pellets per ton, (wrapped in newspaper) in some of the bags. Such storage should not be disturbed frequently for greater effectiveness.



- For fumigants to be effective, they must be applied where there is air-tight storage condition in sacks, containers or buildings.
- For bagged produce, the rate is 1-2 tablets per 100 kg grain sack. The tablet is wrapped in small tissue paper and placed in the middle of the sack. Also, grains must be properly dried; below 12 percent moisture level before storage to avoid mold. Clean granaries and stores before filling with new grains. A farmer can buy these chemicals from reputable agro chemicals dealers or retailers in towns and cities.
- **Note:** Whenever phostoxin tablets are used, ensure the store is airtight to avoid the escape of fumes that are poisonous to human beings and animals when inhaled.

Packaging

While some large farmers in, deliver maize in bulk to the mill, this is not a realistic option for traders buying from small farmers. Bags are required. There are many possible arrangements over bags. Some traders expect the farmer to provide the bag, while others will give the farmer a bag in exchange. Others will pay a higher price for bagged maize. Larger traders may supply bags in advance to farmers, although here they run the risk that farmers may sell their maize to other traders.

Transport

Maize is transported on the shoulder or head, by bicycle, wheelbarrow, ox-cart, car, pick-up, bus or truck. As most farms are not situated right next to a road, either the farmer or the trader will have to transport the maize to the nearest road without using motorized transport. When the trader does this his costs will be increased and the price to the farmer will be lower.

Weight losses The weight the trader sells is unlikely to be exactly the same as the weight he buys. If the time between buying and selling is only a few days the loss may only be small, say one percent or so. But if the trader stores the maize for some time and carries out activities to improve the quality of the maize, then the

difference in weights can be considerable, even as high as ten percent. In the days when all maize was handled by marketing boards and maize was sometimes stored for up to two years, losses on stored maize often reached 30 percent. Reasons for losses in weight are:

Spillage. Spillage can be caused by a variety of factors. Using of old bags with small holes in them or failing to close the bags properly leads to maize falling from the bags. Rough handling, for example by throwing bags from trucks, can make the problem worse.

Theft. Traders who transport bags on the top of buses cannot guard their bags against theft all the time because they are sitting in the body of the bus. Traders who do not accompany their maize bags on trucks run the risk that small quantities will be pilfered, either by the truck driver or by passengers he picks up.

Moisture loss. Maize purchased soon after harvest will probably have a high moisture content and will continue to lose moisture for some time. Depending on the time between purchase and sale this can lead to a noticeable weight loss.

Cleaning. Maize bagged by small farmers often contains straw, stones and other “foreign matter”. Traders selling in local markets cannot sell such maize and have to clean it before sale. This therefore represents another weight loss. Some farmers often try to defraud traders by deliberately placing large stones or other objects in a bag of maize.

Damage in storage. Maize held in storage is vulnerable to attack from insects, rodents and birds, particularly if it is not well fumigated or if it is stored outdoors under tarpaulin. This can lead to significant weight loss when the maize is stored for a long period, for example when the trader is storing in the hope that the price will rise at the end of the marketing year.

Hard facts about weeds in maize

- The annual yield loss in maize as a result of weed problems can reach 50% or more.
- The crop must have a good head start ahead of weeds.
- The crop must be weed free for the first 10 weeks of the crop cycle after crop emergence
- This is the most critical time when the crop requires at least 50% of the nutrients and hence must be weed free.
- If you fail to control weeds in maize in the first 5 weeks of the crop cycle, then 50% of the yields will be lost.
- Never allow weeds to seed, it will enhance the weed seed bank and result in future weed control costs increases and difficulties.

Summary:

- Harvesting should be done when silks has turned brown.
- Post-harvest processing can be either by hand shelling or machines.
- Proper drying helps preserve the life cycle of the grains.
- On-farm sun drying can be used too.
- Storage protects the grains from water and pests.
- Maize can be preserved with phostoxin.

Question and Answer Session

1. When and how should you harvest maize?
2. Should you save your seeds for planting in next season if you planted a hybrid maize?
3. How should you store your maize?
4. What is phostoxin?

Module 9: Record Keeping, Budget Planning and Decision making

Record keeping

Record keeping is the art of collecting useful pieces of data or information on the happenings of a particular undertaking, with the view of processing it in the future (for example, analysing sales and costs and calculating profits). Although record keeping is an important aspect of the family farm business, many farmers do not keep records. This makes it difficult for them to know how much they spent and how much income was earned from their farm business enterprises. Without good farm records, it becomes difficult for farmers to identify problem areas and know whether their businesses are generating profit or not. Therefore, record keeping is an important activity that is necessary for operating farm businesses

successfully. Family farm businesses should keep records, and this responsibility should be shared by all family members. Responsibility for any type of record should be allocated to the family member who is most capable, regardless of their sex.

Why should records be kept in a farm business?

- To track individual enterprise performance.
- To evaluate farm/enterprise profitability.
- To establish a basis for pricing.
- To facilitate access to loans.
- Records are helpful when farmers seek advice from extension agents.
- Records are a good management tool because they provide figures for planning and budgeting.

What is farm budgeting?

A budget is a formal plan for carrying out business activities in the future. It shows the process of carrying out an activity and the end result. Budgeting is the planning process or the development of a plan of action (budget). Families involved with farm businesses should budget together and share responsibilities in the budgeting process.

Uses of farm budgets

Budgets are normally used to:

- Decide what to produce, how much to produce, and the resources needed.
- Itemise the financial aspects of the farming business.
- List the inputs and production practices required by an enterprise.
- Evaluate the performance of different farm enterprises.
- Estimate benefits and costs of changes in production practices.
- Provide the basis for a total farm plan.
- Show the capacity of the farming business to carry risk.
- Support applications for credit (show the ability of the business to pay debt).
- Inform all interested parties of the costs incurred in producing an agricultural product\

Decision making

Decision-making is central to farm management. Each decision has an impact on the farm and on the farm household. Even deciding to do nothing is a decision and has an impact. The more a farmer is aware of the decision-making processes that affect farm and household, the more sustainable the enterprise will be and the more likely it will be profitable and sustainable.

Two of the main features for understanding economic decision-making are:

- The way farm boundaries (decision-making boundaries) are determined.
- The ultimate social objectives for which farm goods are produced.

Production decisions depend on a number of factors that fall outside the physical/ production area of the farm. One of these factors is the farm family/household. In many parts of Nigeria, the farm and the household are virtually one entity. Decisions about the farm directly impact on the household and decisions about the family directly impact on the farm.

Farm boundaries are determined by family structure.

There may be a family farm under the head of household, but the farm may have various sub-units over which a family member will have some level of control. This will have implications for decision-making, particularly with reference to shared labour and equipment. Traditionally men are heads of households, but there are now many variations. Even within a clearly established arrangement, different members of the family, especially women make different decisions for the family at different times of the year.

Key aspects of decision-making

In order to make fundamental decisions farmers need to develop understanding and skills in four broad decision areas:

- **Diagnosis:** Diagnosis means looking at the farm and household as it functioned over some period of time. For a first diagnosis, the farmer may want to understand how the farm has produced over several seasons.
- **Planning:** Once diagnosis is complete, planning can begin; decisions about what, how and how much to produce. Some planning decisions will be based on knowledge, such as how much land and labour are available. Other planning decisions will be based on less certain things, such as rainfall and product prices. The plan will also include an indication of the expected results in terms of yields and income.
- **Implementing**: Implementing means putting the plan into action. In general, implementing does not require major decisions. Things may not always work according to plan; less rainfall than expected, the price of a crop changes etc. During implementation, plans may need to be adjusted to accommodate such changes.
- **Monitoring & Evaluation:** Monitoring means keeping track of what is happening on the farm and the plan is a guide for monitoring. Monitoring provides the farmer with the information needed to evaluate the success of the plan.

Learning how to make decisions in these four areas will put farmers in greater command of the resources and processes that influence their food security and their income generation.

The four areas flow in a pattern which supports continuous learning processes about what works best for the farmer and the farm family.

Planting record sheet

Plant Date	Crop	Variety And the source	Total square meter planted
1/30/12	Cowpea		10000

Monthly Labour/activities records

Date:	Crop:	Bed Number(s):	Activity:	Time (to the15 minute):	Paid labor/unpaid (e.g. Yourself)
EXAMPLE	Maize	12	weeding	15 minutes	

Monthly/Revenue record sheet

	Date	Crop	Harvest amount (ton)	Quantity/ hectare	Quantity sold	Price	Total revenue	NOTES:
Exp:	5/1	Maize	24	12ton/ha	6 ton			
1)								
2)								
3)								
4)								
5)								
6)								
7)								
8)								
9)								
10)								
11)								
12)								
13)								
14)								
15)								
16)								

Keep careful track each month of amounts harvested & how much revenue your selected crops bring you. Use a separate line for each crop and for each harvest date. Use additional sheets if necessary.